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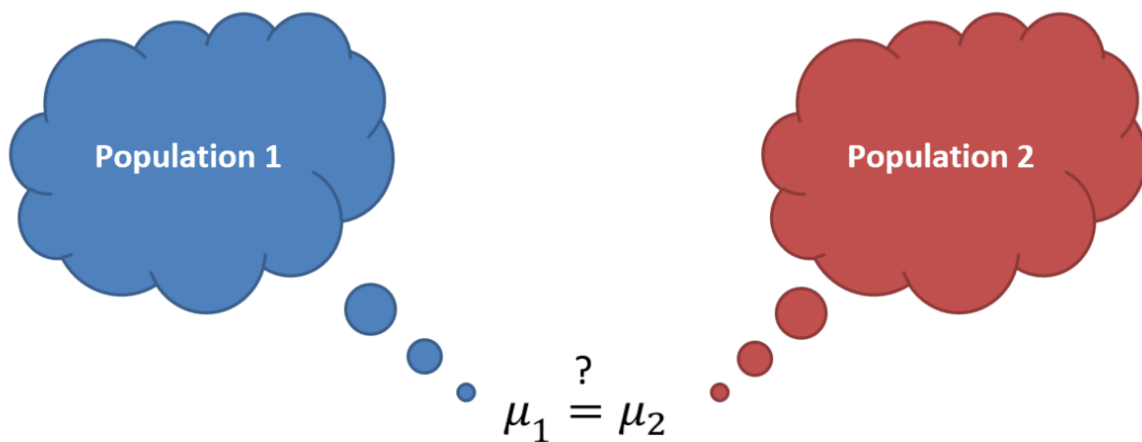
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Litterature:

Introduction to Statistics - Henrik Pedersen

Comparing two population means with known variance



- Fx. The height of people from Fyn (μ_1) and Jylland (μ_2)

Statistical model

- Suppose we have two populations with samples $X_{11}, X_{12}, \dots, X_{1n_1}$ drawn from a normally distributed population

$$X_{1i} \sim N(\mu_1, \sigma_1^2), i = 1, 2, \dots, n_1$$

- and samples $X_{21}, X_{22}, \dots, X_{2n_2}$ drawn from a second normally distributed population

$$X_{2i} \sim N(\mu_2, \sigma_2^2) \quad i = 1, 2, \dots, n_2$$

*The exact models
not important*

- Statistical question of interest:

$$\mu_1 = \mu_2?$$

- Note that in general, the two samples are of different size (i.e., $n_1 \neq n_2$).

Comparing two population means with known variance

Parameter Estimate

- From the central limit theorem, we know that the estimates of the two population means are

$$\hat{\mu}_1 = \bar{x}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} X_{1i} \sim N(\mu_1, \sigma_1^2/n_1)$$

$$\hat{\mu}_2 = \bar{x}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} X_{2i} \sim N(\mu_2, \sigma_2^2/n_2)$$

- The estimate of the difference between the two population means is

$$\bar{x}_1 - \bar{x}_2 \sim N(\mu_1 - \mu_2, \sigma_1^2/n_1 + \sigma_2^2/n_2)$$

- (the sum of two Gaussian PDFs is another Gaussian).

$$\text{Mean of } (\bar{X}_1 - \bar{X}_2) = E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$$

$$\text{Variance of } (\bar{X}_1 \pm \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2) \pm 2 \text{Cov}(\bar{X}_1, \bar{X}_2)$$

$$\text{Assuming independence: } \text{Cov}(\bar{X}_1, \bar{X}_2) = 0$$

$$\Rightarrow \text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

Comparing two population means with known variance

Hypothesis Test for Comparing Two Population Means

(known σ 's)

- Here we will assume that the variances of the two populations are equal

$$\sigma^2 = \sigma_1^2 = \sigma_2^2$$

- Under the null hypothesis $H_0: \delta = \mu_1 - \mu_2 = 0$, we must have

$$\bar{x}_1 - \bar{x}_2 \sim N(0, \sigma^2/n_1 + \sigma^2/n_2)$$

- Standardizing, we get

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\text{Var}(\bar{x}_1 - \bar{x}_2)}} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\sigma^2/n_1 + \sigma^2/n_2}} = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{1/n_1 + 1/n_2}} \sim N(0,1)$$

Denominator is called the **standard error** of $\bar{x}_1 - \bar{x}_2$ (**SE**):

Remember

$$Z = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}$$

Comparing two population means with known variance

Example: Two-sample z-test (known σ 's)

Given

- $\bar{x}_1 = 3, \bar{x}_2 = 4, \sigma_1 = \sigma_2 = 1, n_1 = 10, n_2 = 20$
- $H_0 : \mu_1 - \mu_2 = 0$ (two-sided), significance level $\alpha = 0.05$

Test statistic

$$z = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sigma \sqrt{1/n_1 + 1/n_2}} = \frac{3 - 4}{1 \sqrt{1/10 + 1/20}} = \frac{-1}{\sqrt{0.15}} = -2.582.$$

Under $H_0, z \sim N(0, 1)$.

Method A: Critical-value (rejection region)

- Two-sided test at $\alpha = 0.05 \rightarrow$ critical values

$$z_{\alpha/2} = \Phi^{-1}(1 - \frac{\alpha}{2}) = z_{0.975} \approx 1.96$$

- **Rejection region:** $|z| > 1.96$.
- **Observed:** $|-2.582| = 2.582 > 1.96 \rightarrow$ **Reject H_0 .**

(At $\alpha = 0.01, z_{0.995} \approx 2.576$; since $2.582 > 2.576$, we'd also reject at 1%.)

Python:

```
z_crit = norm.ppf(0.975)
```

Method B: p-value

- Two-sided p-value:

$$p = 2(1 - \Phi(|z|)) = 2(1 - \Phi(2.582)) \approx 2(1 - 0.9951) = 0.0098.$$

- Compare to $\alpha = 0.05$: $p = 0.0098 < 0.05 \rightarrow$ **Reject H_0 .**
(Also $p < 0.01 \rightarrow$ reject at 1%.)

Python:

```
p_two_sided = 2 * (1 - norm.cdf(abs(z)))
```

Comparing two population means with known variance

95 % Confidence Interval for the Difference Between Two Population Means (known σ' s)

True difference

$$\delta = \mu_1 - \mu_2$$

Estimator (sample-based estimate)

$$\hat{\delta} = \bar{X}_1 - \bar{X}_2 \quad \text{with} \quad \hat{\delta} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Confidence level condition

We seek limits δ_- and δ_+ such that the true difference δ lies within them with 95% probability:

$$\Pr(\delta_- \leq \delta \leq \delta_+) = 0.95$$

Standardizing

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - \delta}{\sigma \sqrt{1/n_1 + 1/n_2}} \sim N(0, 1)$$

For a 95% confidence level,

$$0.95 = \Pr(-1.96 \leq Z \leq 1.96)$$

Substituting the standardized form gives

$$0.95 = \Pr\left(-1.96 \leq \frac{(\bar{X}_1 - \bar{X}_2) - \delta}{\sigma \sqrt{1/n_1 + 1/n_2}} \leq 1.96\right)$$

Comparing two population means with known variance

95 % Confidence Interval for the Difference Between Two Population Means (known σ 's)

Isolating δ

$$0.95 = \Pr\left((\bar{X}_1 - \bar{X}_2) - 1.96 \sigma \sqrt{1/n_1 + 1/n_2} \leq \delta \leq (\bar{X}_1 - \bar{X}_2) + 1.96 \sigma \sqrt{1/n_1 + 1/n_2}\right)$$

Endpoints of the 95% confidence interval

$$\begin{aligned} \delta_- &= (\bar{X}_1 - \bar{X}_2) - 1.96 \sigma \sqrt{1/n_1 + 1/n_2} \\ \delta_+ &= (\bar{X}_1 - \bar{X}_2) + 1.96 \sigma \sqrt{1/n_1 + 1/n_2} \end{aligned}$$

Or equivalently, in compact notation:

$$\hat{\delta} \pm 1.96 \sqrt{\text{Var}(\hat{\delta})} \quad \text{where } \hat{\delta} = \bar{X}_1 - \bar{X}_2$$

Interpretation

We are **95% confident** that the true difference in population means, $\delta = \mu_1 - \mu_2$, lies within this interval centered at the observed difference $\hat{\delta}$.

Comparing two population means with known variance

Example: Two-sample z-test (known σ 's)

Given

$$\bullet \bar{x}_1 = 3, \bar{x}_2 = 4, \sigma_1 = \sigma_2 = 1, n_1 = 10, n_2 = 20$$

Confidence interval formula

$$\delta_{\pm} = (\bar{X}_1 - \bar{X}_2) \pm 1.96 \cdot \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Lower bound

$$\begin{aligned} \delta_- &= (\bar{X}_1 - \bar{X}_2) - 1.96 \cdot \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ &= (3 - 4) - 1.96 \cdot 1 \cdot \sqrt{\frac{1}{10} + \frac{1}{20}} \\ &= -1 - 1.96 \cdot \sqrt{0.15} \\ &= -1 - 1.96(0.3873) = \boxed{-1.759}. \end{aligned}$$

Upper bound

$$\begin{aligned} \delta_+ &= (\bar{X}_1 - \bar{X}_2) + 1.96 \cdot \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ &= (3 - 4) + 1.96 \cdot 1 \cdot \sqrt{\frac{1}{10} + \frac{1}{20}} \\ &= -1 + 1.96(0.3873) = \boxed{-0.241}. \end{aligned}$$

Since $\delta = 0$ is **not contained** within this 95% confidence interval, we reject the null hypothesis

$$H_0 : \mu_1 - \mu_2 = 0$$

and conclude that the two population means are significantly different.

Comparing two population means with unknown variance

What changes when the variance is unknown?

Now σ^2 is unknown.

So we cannot compute

$$\frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

We **assume** both populations have the **same variance**, even though we do not know its value.

$$\sigma_1^2 = \sigma_2^2 = \sigma^2.$$

So instead of estimating two separate variances, we combine information from both samples to estimate the one common variance.

That combined estimator is the **pooled variance**.

Comparing two population means with unknown variance

Sample variance $\Rightarrow$ pooled variance

Start with the unbiased estimators:

$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1)^2, \quad s_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (x_{2i} - \bar{x}_2)^2.$$

The corresponding **total sums of squared deviations** are:

$$S_1 = (n_1 - 1)s_1^2, \quad S_2 = (n_2 - 1)s_2^2.$$

To get one combined estimate, add these and divide by the **total degrees of freedom**:

$$s_p^2 = \frac{S_1 + S_2}{(n_1 - 1) + (n_2 - 1)} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

NB. We cannot calculate $s_1^2 + s_2^2$ directly

Because:

- s_1^2 and s_2^2 are already **averages**
- You cannot add averages directly without accounting for sample sizes

That would ignore how much data each variance is based on.

Comparing two population means with unknown variance

Pooled variance is an unbiased estimator under equal-variance assumption

Using linearity of expectation,

$$\begin{aligned} E[s_p^2] &= E\left[\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right] \\ &= \frac{(n_1 - 1)E[s_1^2] + (n_2 - 1)E[s_2^2]}{n_1 + n_2 - 2}. \end{aligned}$$

Under equal variance,

$$E[s_1^2] = E[s_2^2] = \sigma^2.$$

So

$$\begin{aligned} E[s_p^2] &= \frac{(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2}{n_1 + n_2 - 2} \\ &= \frac{(n_1 + n_2 - 2)\sigma^2}{n_1 + n_2 - 2} = \sigma^2. \end{aligned}$$

So s_p^2 is an unbiased estimator of the common variance.

Comparing two population means with unknown variance

T-test (Unknown Variances)

We have seen that

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\text{Var}(\bar{x}_1 - \bar{x}_2)}} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\sigma^2/n_1 + \sigma^2/n_2}} = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{1/n_1 + 1/n_2}} \sim N(0,1)$$

Replace σ with the **pooled** standard deviation

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Then we get the the **t-statistic**:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

Denominator is called the **standard error** of the difference of sample means (**SE**):

$$\text{SE}(\bar{X}_1 - \bar{X}_2) = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

Comparing two population means with unknown variance

Example: Hypothesis Test for Comparing Two Population Means (pooled variance)

Given

- $\bar{x}_1 = 3, \bar{x}_2 = 4$
- $n_1 = 10, n_2 = 20$
- $s_1^2 = 1.4, s_2^2 = 1.1$
- Two-sided test of $H_0 : \delta = \mu_1 - \mu_2 = 0$ at $\alpha = 0.05$
- **Equal-variance** model (so we pool)

Pooled variance

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(10 - 1) \cdot 1.4 + (20 - 1) \cdot 1.1}{10 + 20 - 2}$$

$$= \frac{9 \cdot 1.4 + 19 \cdot 1.1}{28} = \frac{12.6 + 20.9}{28} = \frac{33.5}{28} = 1.1964,$$

$$s_p = \sqrt{1.1964} = 1.0938,$$

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 1.0938 \sqrt{\frac{1}{10} + \frac{1}{20}} = 1.0938 \sqrt{0.1 + 0.05}$$

$$= 1.0938 \sqrt{0.15} = 1.0938 \cdot 0.3872983346 \approx 0.4237.$$

Comparing two population means with unknown variance

Example: Hypothesis Test for Comparing Two Population Means (pooled variance)

Test statistic and df

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE} = \frac{(3 - 4) - 0}{0.4237} = \frac{-1}{0.4237} = -2.3606$$

Degrees of freedom:

$$\nu = n_1 + n_2 - 2 = 10 + 20 - 2 = 28.$$

Method A: Critical value

Two-sided critical value at $\alpha = 0.05$:

$$t_{0.975, 28} = t_{inv}(0.975, 28) \approx 2.048$$

Decision:

$$|t| = 2.3606 > 2.048 \Rightarrow \boxed{\text{Reject } H_0}.$$

Python: `scipy.stats.t.ppf(0.975, 28)`

Method B: p-value

Two-sided p-value using the Student-t CDF:

$$p = 2 \left(1 - t_{\text{cdf}}(|t|, \nu) \right) = 2 \left(1 - t_{\text{cdf}}(2.3606, 28) \right).$$

From tables/software:

$$t_{\text{cdf}}(2.3606, 28) \approx 0.9873 \Rightarrow p = 2(1 - 0.9873) = 2(0.0127) = \boxed{0.0254}.$$

Compare with $\alpha = 0.05$:

$$p = 0.0254 < 0.05 \Rightarrow \boxed{\text{Reject } H_0}.$$

Python: `p = 2 * (1 - scipy.stats.t.cdf(abs(2.3606), 28))`

Comparing two population means with unknown variance

Example: Hypothesis Test for Comparing Two Population Means (pooled variance)

Compute the confidence interval limits

The general formula:

$$\delta_{\pm} = (\bar{x}_1 - \bar{x}_2) \pm t_{\text{inv}}(0.975, n_1 + n_2 - 2) \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Lower bound:

$$\begin{aligned}\delta_- &= (3 - 4) - t_{\text{inv}}(0.975, 28) \cdot 1.0938 \cdot \sqrt{\frac{1}{10} + \frac{1}{20}} \\ &= -1 - 2.0484 \cdot 1.0938 \cdot 0.3873 \\ &= -1 - 2.0484 \cdot 0.4236 \\ &= -1 - 0.8678 = \boxed{-1.8678}\end{aligned}$$

Upper bound:

$$\begin{aligned}\delta_+ &= (3 - 4) + t_{\text{inv}}(0.975, 28) \cdot 1.0938 \cdot \sqrt{\frac{1}{10} + \frac{1}{20}} \\ &= -1 + 2.0484 \cdot 1.0938 \cdot 0.3873 \\ &= -1 + 2.0484 \cdot 0.4236 \\ &= -1 + 0.8678 = \boxed{-0.1322}\end{aligned}$$

Since $\delta = 0$ is **not included** in the 95% confidence interval, we **reject the null hypothesis**

$$H_0 : \mu_1 - \mu_2 = 0$$

and conclude that the two population means are **significantly different** at the 5% significance level.

Paired vs unpaired tests

So far we have looked at **unpaired (independent) tests**. Now we consider **paired (dependent) tests**.

Unpaired Test (Independent Samples)

Used when comparing **two separate groups**, where observations in one group are **independent** of those in the other.

We compare the **group means**.

Paired Test (Dependent Samples)

Used when the **same subjects** (or closely matched subjects) are measured twice, for example:

- before and after a treatment
- under two different conditions

We analyze the **differences within each pair**.

When to Use Which?

- If you can measure the **same individual twice**, a **paired test** is often preferred, because it removes variability due to differences between individuals.
- If this is not possible, you compare **two independent groups** using an unpaired test.

Paired tests

We have two related measurements on the same subjects (or matched pairs):

$$X_{1i}, X_{2i}, \quad i = 1, 2, \dots, n$$

We define the **true mean difference** between the two populations as

$$\delta = \mu_1 - \mu_2$$

Assumptions

- Each *pair* (X_{1i}, X_{2i}) comes from the same subject or matched unit.
- The **pairs are independent of each other**, meaning that what happens for subject i does not influence subject j for $i \neq j$.
- However, the **two measurements within the same pair** $(X_{1i}$ and $X_{2i})$ may be **correlated** — for example, a before/after measurement on the same person is likely related.

Paired tests

Paired tests

Define the within-pair differences

$$D_i = X_{1i} - X_{2i}, \quad i = 1, 2, \dots, n.$$

Each D_i represents how much the two measurements differ for one subject.

Note on covariance

The two measurements within the same pair, X_{1i} and X_{2i} , may be correlated. This is handled automatically when working with the differences, since

$$\text{Var}(D_i) = \text{Var}(X_{1i}) + \text{Var}(X_{2i}) - 2 \text{Cov}(X_{1i}, X_{2i}).$$

If the covariance is positive (as in most before/after settings), the variance of D_i is reduced. This is why paired tests are often more efficient than unpaired tests.

We assume that the differences $D_1, \dots, D_n$ are independent and identically distributed with

$$\mathbb{E}[D_i] = \delta, \quad \text{Var}(D_i) = \sigma^2.$$

The sample mean difference is

$$\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i.$$

This is an unbiased estimator of δ .

Paired tests

By the Central Limit Theorem (for sufficiently large n),

$$\bar{D} \approx \mathcal{N}\left(\delta, \frac{\sigma^2}{n}\right).$$

The sample variance of the differences is

$$s_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2.$$

Remark (small samples)

If n is small, the normal approximation from the Central Limit Theorem may be inaccurate. In that case, one should either check that the differences are **approximately normally distributed**.

Paired tests

Because $\bar{D} \sim N(\delta, \sigma^2/n)$,

$$Z = \frac{\bar{D} - \delta_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Remember

$$Z = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}$$

- $\bar{D}$ is your estimator of the true mean difference δ
- Under the null hypothesis, we assume:

$$\delta = \delta_0$$

We **don't know** σ , the true standard deviation of the differences.

We only have an **estimate** of it from our data:

$$s_D = \sqrt{\frac{1}{n-1} \sum (D_i - \bar{D})^2}.$$

So we have to **replace** σ by s_D in our formula.

This gives the standardized test statistic

$$t = \frac{\bar{D} - \delta_0}{s_D/\sqrt{n}} \sim t_{n-1}.$$

Paired tests

Example: Corn Growth

A farmer decides to try out a new fertiliser on a test plot containing 10 stalks of corn.

Before applying the fertiliser, he measures the height of each stalk.

Two weeks later, he measures the stalks again, being careful to match each stalk's new height to its previous one.

The stalks would have grown an average of 6 inches during that time even without the fertiliser.

Did the fertiliser **change the expected mean**?

Stalk	1	2	3	4	5	6	7	8	9	10
Before height	35.5	31.7	31.2	36.3	22.8	28.0	24.6	26.1	34.5	27.7
After height	45.3	36.0	38.6	44.7	31.4	33.5	28.8	35.8	42.9	35.0

Define hypotheses

We test whether fertilizer **increases** growth:

$$H_0 : \delta = 6 \quad (\text{no extra effect})$$

$$H_1 : \delta > 6 \quad (\text{fertilizer increases growth})$$

where δ is the true mean increase in height.

Paired tests

Example: Corn Growth

Compute the paired differences

Stalk	D_i	$(D_i - \bar{D})^2$
1	9.8	5.954
2	4.3	9.364
3	7.4	0.002
4	8.4	1.082
5	8.6	1.538
6	5.5	3.460
7	4.2	9.986
8	9.7	5.476
9	8.4	1.082
10	7.3	0.004

The sample mean difference is:

$$\bar{D} = \frac{1}{10} \sum_{i=1}^{10} D_i = 7.36$$

The sample variance is:

$$s_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2 = 4.216 \Rightarrow s_D = \sqrt{4.216} = 2.05$$

Paired tests

Example: Corn Growth

We use the one-sample t -statistic for paired data:

$$t = \frac{\bar{D} - \delta_0}{s_D / \sqrt{n}}$$

Substitute the values:

$$t = \frac{7.36 - 6}{2.05 / \sqrt{10}} = \frac{1.36}{0.648} = 2.09$$

Method A: Critical value (rejection region)

- One-sided test at $\alpha = 0.05$, degrees of freedom $\nu = 9$:

$$t_{\text{crit}} = t_{\text{inv}}(0.95, 9) \approx 1.833.$$

- Decision rule:** Reject H_0 if $t > t_{\text{inv}}(0.95, 9)$.
- Observed:** $t = 2.09 > 1.833 \Rightarrow \boxed{\text{Reject } H_0}$.

Python: `t_crit = t.ppf(0.95, df=9)`

Method B: p-value

- One-sided p-value for $H_1 : \delta > 6$:

$$p = 1 - t_{\text{cdf}}(t, \nu) = 1 - t_{\text{cdf}}(2.09, 9) \approx 1 - 0.9669 = \boxed{0.0331}.$$

- Compare with $\alpha = 0.05$: $p = 0.0331 < 0.05 \Rightarrow \boxed{\text{Reject } H_0}$.

Python: `p_one_sided = 1 - t.cdf(t_stat, df=9)`